Summer 2025 Course Syllabus
WB11 Section

INST327 - Database Design and Modeling

Course Description and Learning Outcomes

This course is a broad introduction to relational database systems, combining both conceptual understanding and hands-on practice. Students will explore the relational model, which forms the foundation for designing and querying databases. Key topics include database design techniques such as user-focused design, requirements analysis, entity relationship modeling, and normalization.

Through practical exercises, students will use Structured Query Language (SQL) and a database management system (DBMS) to build, populate, and query a functioning database. Along the way, students will also use Excel for preparing data and explore platforms such as Kaggle, Google Dataset Search, and Data.gov to find and work with datasets.

By the end of this course, you will be able to:

- Explain the relational model as a logical framework for organizing and retrieving data.
- Write database queries using SQL.
- Translate user needs into clear database requirements and represent them using entity-relationship models.
- Design and build a working relational database using a DBMS.
- Normalize and de-normalize a database to improve performance.
- Recognize potential security issues in databases and develop strategies to address them.
- Understand how the design and use of relational databases connect to broader social and organizational structures, including ethical and equity considerations.

Instructional Team

Iskander Lou
Lecturer
ilou@umd.edu
(please use Canvas/Discord for communication)

Nicolas Lizarralde
Graduate TA

Elizabeth Vandegrift
Undergraduate TA

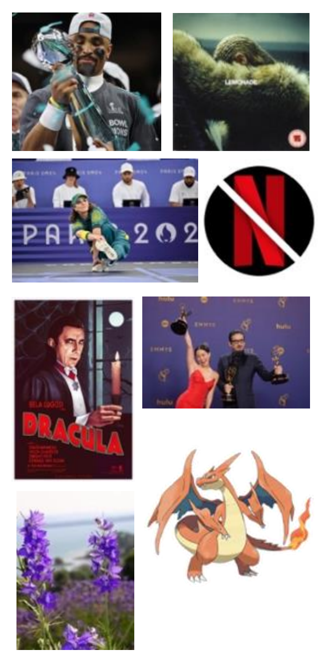
Office Hours

The office hours schedule is available in the course homepage.

Class Time

Mon & Wed	1pm-3:15pm EST	Location: Zoom

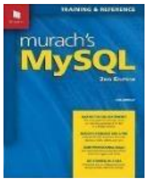
Databases that we have built before



(Clues are provided in the last page of the syllabus.)

Required Resources

Textbook



Murach's MySQL, Third Edition (2019)

Murach, Joel. ISBN: 9781943872367

Physical copies are available at campus bookstore.
eBook: https://www.murach.com/shop/murach-s-mysql-3rd-edition-detail
Chapters 2 & 3 can be downloaded for free at the link above.
Additional readings and resources may be assigned via ELMS.

Software

MySQL Community Server (v8 or later) ← *install this first*

- Mac: https://dev.mysql.com/downloads/mysql/
- Windows: https://dev.mysql.com/downloads/installer/

MySQL Workbench (v8 or later) ← *install this after*

- Mac: https://dev.mysql.com/downloads/workbench/
- Windows: The community server installer includes Workbench as well.

Excel

- Mac: https://terpware.umd.edu/Mac/Title/3107
- Windows: https://terpware.umd.edu/Windows/Title/3107

Useful Links

Excalidraw (easy way to draw and share diagrams)
https://excalidraw.com/

Kaggle (the most popular platform for datasets)
https://www.kaggle.com/

Google Dataset Search (search engine for discovering datasets across the web)
https://datasetsearch.research.google.com/

Data.gov (public datasets provided by the U.S. government)
https://data.gov/

We will work through SQL examples during synchronous class meetings. Please have the required software installed before or during the first class. Check ELMS for download and installation instructions, videos, and other resources. Feel free to contact the instructional team if you need assistance with software installation.

Campus Policies

Please visit www.ugst.umd.edu/courserelatedpolicies.html for the Office of Undergraduate Studies' full list of campus-wide policies (regarding academic integrity, student privacy, non-discrimination, accessibility, etc.) and follow up with me if you have any questions.

Course Policies

- **Monitor ELMS announcements regularly**; adjust settings as needed to ensure receipt of all notifications.
- Follow course calendar activities:
    - Watch the week's assigned videos, keep up with readings, and execute examples before the lecture.
    - Complete project, homework, and lab assignments as assigned.
    - Complete lab assignments during discussion session (or soon after) and submit via ELMS.
    - Submit all individual and team project assignments via ELMS on time. Some assignments cannot be submitted late, and others have late penalties.
- **Discussion attendance is required** to receive credit for the weekly lab assignments**.**
- The office hours schedule will be posted on the course homepage in ELMS. If you are struggling with course content, please reach out to the instructional team via **Discord or visit office hours**.
- **All team members are expected** to actively participate in the team project**.** At the end of the course, students will complete a survey detailing their contributions. A lack of meaningful contribution may result in a grade of zero for the project.
- **Homework assignments must be your own work**. Obtaining answers from online answer providers, AI tools, or any individual other than the instructional team is a violation of course policy and may result in a referral to the Office of Student Conduct.
- If you would like to request an excused absence or an assignment due date extension due to a university-sanctioned reason, please fill out the **Accommodation Request Form** (available in course homepage). If you don't hear back from us within two business days, please send a follow-up message via Canvas.
- If you qualify for ADS accommodations, please notify the instructor as soon as possible.

Course Structure

The course is delivered in a hybrid learning format structured around weekly modules. As such, there are asynchronous elements and synchronous classes.

Self-study Preparation (Asynchronous Materials): This is a combination of readings, slides, and coding examples. Students are expected to complete the assigned readings and coding examples prior to the relevant lecture class.

Lecture and Discussion Classes (Synchronous Sessions): Lectures and labs will be interactive and based on the material covered in the *Self-Study Preparation*. Participation is necessary to excel at this course. If you anticipate missing a session, please submit the accommodation request form ahead of time, or soon after missing the class.

Course Activities and Assessments

This course involves formative and summative assessment components, as well as an applied team project, which serves as the Scholarship in Practice component for the course. Many of the activities and assessments are designed to support the semester-long Team Database Design project. Activities gauge your understanding and expertise of the SQL skill set and readiness to perform the related project work.

Quizzes: Quizzes will assess your understanding of the concepts and skills covered in the preceding module. They are to be completed individually and will be available online in ELMS, with a 60-minute time limit.

Labs: Lab assignments provide practice of the skills necessary for a comprehensive understanding of SQL and the successful completion of the team project. You will receive these practice problems on Monday and Wednesday morning, and you must complete them during class with the TA's assistance as needed. You will submit your work individually via ELMS. Discussion attendance is required to receive credit for the lab assignment.

Homework Assignments: There will be several homework assignments, each of which will include multiple questions. The questions will be practical tasks, such as writing SQL queries, constructing a view, normalizing a table, or developing a stored procedure. The assignments are individual work. Although you may consult with your classmates, TAs, and the instructor to develop general approaches to solving questions, you must work individually while you build, type, test, and debug your answers. You are not allowed to share SQL, Excel, or diagram files with classmates while completing homework assignments.

Team Project: Students will work in **3 person groups** to design and build a non-trivial relational database. As a Scholarship in Practice course, project-related work is central to this course. The project will involve defining the team structure and responsibilities for individual team members, identifying an information need for a relational database, determining the requirements for the database, developing a deadline-oriented plan and schedule for building the database, designing the logical specifications, building, and populating the database, and developing queries/views that will showcase the capabilities of the database for fulfilling the identified user needs. Students will form their teams in week 1 or be assigned to a team by the instructor. Teams will choose their own topic. Central to this project will be team management and team interaction. Project deliverables throughout the semester will include assessment of team function as well as the software development process. The TAs will act as team mentors throughout the life cycle of the project. Project teams will provide periodic updates about their progress on the project. Details will be made available during the semester.

Grades

Grades will be posted on ELMS. If you would like to discuss your grade or have questions about how something was scored, please contact the instructional team. Grade disputes must be submitted in writing within one week of receiving the grade.

Late submissions won't be accepted. If you need an extension, please submit the Accommodation Form ahead of the deadline. Requests sent after the due date are less likely to be approved.

The table below illustrates the percentage weight of each assessment towards your final grade

Component		Percentage
Quizzes		10%
Labs (required attendance)		20%
Homework Assignments		40%
Team Project		30%
Project Proposal	5%	
Project Standups (required attendance)	5%	
Progress Report	5%	
Final Project Report	15%	

Letter grades will be assigned using the following scale.

Grading Scale				
A+ ≥ 97%	B+ ≥ 87.00%	C+ ≥ 77.00%	D+ ≥ 67.00%	F <60.0%
A ≥ 93.00%	B ≥ 83.00%	C ≥ 73.00%	D ≥ 63.00%	
A- ≥ 90.00%	B- ≥ 80.00%	C- ≥ 70.00%	D- ≥ 60.00%	

Tentative Course Schedule

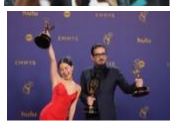
This schedule is for planning purposes and may change. See ELMS for current information.

Week	Date	Topics	Concepts	Assignments Unlock
1	Monday June 2, 2025	Course Introduction Simple SELECT Queries	Learning agendas Logistics Software installation SELECT query syntax: select statement, table selection, data filtering, ordering, aliasing, and concatenating	Lab 1 Quiz 1
1	Wednesday June 4, 2025	Multi-Table SELECT Queries	Workbench Navigation JOINs: inner, left, right, outer UNION	Lab 2 Quiz 2 Homework 1 (SELECT queries, JOINs, UNION)
2	Monday June 9, 2025	Database Design Part One	Entities & attributes Entity relationships	Lab 3 Quiz 3 Project Proposal
2	Wednesday June 11, 2025	Database Design Part Two	Three Normal Forms 1NF, 2NF, 3NF	Lab 4 Homework 2 (Database Design)
3	Monday June 16, 2025	Data Preparation in Excel	Excel tips and shortcuts Delimiters, duplicates, blank rows XLOOKUP	Lab 5 Progress Report

3	Wednesday June 18, 2025	Database Development	Entity-Relationship Diagram (ERD) Forward engineering Data import Database export	Lab 6 Homework 3 (Database Development) Project Standup 1
4	Monday June 23, 2025	Aggregate Functions	AVG, MAX, MIN, COUNT, SUM GROUP BY HAVING vs WHERE IF statement CASE expression	Lab 7 Quiz 4
4	Wednesday June 25, 2025	Complex Queries	Subqueries Common Table Expressions (CTE) Ranking functions	Lab 8 Quiz 5 Homework 4 (Complex Queries) Project Standup 2
5	Monday June 30, 2025	Stored Programs	Views Procedures Functions Triggers	Lab 9 Quiz 6 Final Project
5	Wednesday July 2, 2025	More SQL Functions	Trim, case, round, truncate, date format	Lab 10 Quiz 7

6	Monday July 7, 2025	CRUD	Create, alter, drop table Create a copy of a table Create, alter, drop database Insert, update, delete rows	Lab 11
6	Wednesday July 9, 2025	SQL Interview Prep	Practice interview questions from Amazon, Uber, Robinhood, Capital One	Lab 12 (Extra Credit) Project Standup 3

Super Bowl



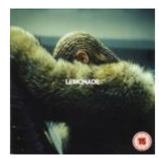
Emmy Awards 2024

Pokémon

Horror movies

Maryland native plants

Paris 2024 Olympics



Greatest albums of all time (according to Rolling Stone magazine)

Cancelled Netflix Series